

Nested RICS-FDFM

Multiscale Information-Geometric Stochastic Hybrid Network Model

*Recursive information-container dynamics with adaptive topology, endogenous hierarchy, and
path-dependent memory*

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Abstract

This paper defines RICS-FDFM as a recursive, multiscale, information-geometric stochastic hybrid system on an adaptive directed hypergraph. Each system carries three probability-valued signal states - official, operational, and meta-signal - together with feedback permeability, fragmentation, latent stress, usable energy, rigidity, adaptive capacity, and explicit memory coordinates. Signal disagreement is measured on the probability simplex by Jensen-Shannon and Fisher-Rao geometry rather than by Euclidean distance alone. Network topology is endogenous: edges can strengthen, decay, appear, disappear, split, merge, and form higher-order coalitions. Higher-level systems are generated recursively from lower-level systems through coarse-graining maps, while downward constraints feed back into lower-level dynamics. Path dependence is represented through Volterra memory kernels and a finite-dimensional Markovian embedding. Continuous dynamics are coupled to state-dependent jump events for vulnerability, collapse, disintegration, decoupling, restoration, node birth/death, and structural reorganization. The resulting model is simulable, estimable, falsifiable, and extensible across computational, biological, psychological, institutional, and socio-technical domains without asserting identical causal parameters across domains.

Revision Scope

- Introduces genuine information-theoretic variables and information geometry.
- Replaces Euclidean signal mismatch with distributional divergence and geodesic distance.
- Makes network topology adaptive, including edge birth/death, coalition formation, and node split/merge events.
- Adds a recursive hierarchy in which systems at level k are generated from systems at level $k-1$ and exert downward constraints.
- Adds explicit memory through convolution kernels, delays, and a Markovian auxiliary-state representation.

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1. Modeling Objective

RICS-FDFM formalizes a recurrent pattern in adaptive systems: declared state, realized operation, and internal self-model can diverge; corrective information can be filtered or suppressed; unresolved divergence can accumulate as structural fragmentation and latent stress; resource exchange can become extractive or destabilizing; and apparently stable output can coexist with rising failure risk. The converse is equally important: a visibly perturbed system may remain recoverable when it preserves information flow, adaptive capacity, and sufficient reserve.

The formal object is not a list of categorical system types. It is a multiscale stochastic hybrid process whose qualitative regimes are measurable regions of a continuous state space and whose discrete failures or reorganizations are state-dependent jumps.

2. Probability Space and Recursive Architecture

Let levels be indexed by $k = 0, 1, \dots, K$. At level k , the active system set is $V^{(k)}(t) = \{1, \dots, N_k(t)\}$. The number of systems may vary because split, merge, birth, and removal events are allowed. All processes are defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$ satisfying the usual conditions.

$$Z(t) = \{X_i^{(k)}(t), R_i^{(k)}(t), W^{(k)}(t), H^{(k)}(t), C^{(k)}(t)\}_{k=0}^K \quad (1)$$

Here X is the continuous state, R the discrete regime, W the directed pairwise network, H the higher-order interaction tensor or hypergraph, and C the interlevel membership/coarse-graining structure. The full process is recursively coupled: lower-level states generate higher-level states, and higher-level states impose downward constraints on their constituents.

3. Information-Valued State Space

Each system i at level k has state

$$X_i^{(k)} = (O_i, P_i, M_i, f_i, s_i, \ell_i, e_i, r_i, a_i, \mu_i, q_i) \quad (2)$$

The first three coordinates are probability distributions on a finite or continuous message space A . For a finite alphabet of size m , O_i, P_i, M_i belong to the interior of the simplex $\Delta^{(m-1)}$. O_i is the official or declared distribution, P_i the operational or realized distribution, and M_i the meta-distribution encoding the system's internal model of the relation between declared and realized behavior.

Symbol	Meaning	Range
O_i	official signal distribution	interior of probability simplex
P_i	operational signal distribution	interior of probability simplex
M_i	meta-signal distribution	interior of probability simplex
f_i	feedback permeability	$(0,1)$
s_i	structural fragmentation	nonnegative
ℓ_i	latent stress load	nonnegative

e_i	usable reserve / adaptive energy	nonnegative
r_i	rigidity / form dominance	(0,1)
a_i	adaptive capacity	(0,1)
μ_i	memory vector / history embedding	$\mathbb{R}^J$ or nonnegative cone
q_i	output or functional performance	application-specific

4. Information Geometry and Alignment

4.1 Distributional signal divergence

The signal layers are not points in an unconstrained Euclidean vector space. They are probability distributions. A symmetric, finite divergence is therefore defined using Jensen-Shannon divergence:

$$JS(U,V) = 1/2 KL(U \parallel (U+V)/2) + 1/2 KL(V \parallel (U+V)/2) \quad (3)$$

with $KL(U \parallel V) = \sum_a U(a) \log[U(a)/V(a)]$. The system-level divergence functional is

$$d_i = \alpha_{OP} JS(O_i, P_i) + \alpha_{PM} JS(P_i, M_i) + \alpha_{OM} JS(O_i, M_i) \quad (4)$$

where all α coefficients are nonnegative. This quantity is bounded for discrete distributions and remains meaningful when supports partially differ.

4.2 Fisher-Rao geometry

For local dynamics and geodesic motion on the simplex, use the Fisher information metric. Under the square-root map $z_a = 2\sqrt{p_a}$, the simplex is embedded on the positive orthant of a sphere. The Fisher-Rao distance is

$$d_{FR}(U,V) = 2 \arccos(\sum_a \sqrt{U(a)V(a)}) \quad (5)$$

The normalized alignment score combines global divergence and local geometry:

$$A_i = \exp[-d_i/d_0] \cdot \exp[-\eta_{FR} d_{FR}(P_i, M_i)^2] \in (0,1] \quad (6)$$

4.3 Information production, surprise, and retained predictive information

Let $Y_i(t)$ be observed outcomes generated conditionally on P_i . Define instantaneous Bayesian surprise and predictive information by

$$Srp_i(t) = KL(\text{Posterior}_i(t) \parallel \text{Prior}_i(t)) \quad (7)$$

$$I_i^{\wedge pred}(t;\tau) = I(Y_i(t); Y_i(t+\tau) \mid F_t) \quad (8)$$

Feedback is useful when surprise is admitted and converted into improved prediction. The effective information assimilation rate is

$$G_i^{\text{info}} = f_i \cdot \text{Srp}_i \cdot [\Delta I_i^{\text{pred}}]_+ \quad (9)$$

This prevents the word information from functioning merely as a metaphor: information enters the state equations through measurable entropy, divergence, surprise, and predictive gain.

5. Memory and Path Dependence

5.1 Volterra memory

History enters the drift through causal convolution operators. For any state variable x_i , define

$$M_i^x(t) = \int_0^t K_x(t-s) x_i(s) ds \quad (10)$$

where K_x is a nonnegative or signed memory kernel. Exponential kernels describe fading memory; power-law kernels describe long memory and can approximate fractional dynamics.

5.2 Finite-dimensional Markovian embedding

To preserve simulation efficiency, approximate each kernel by a Prony series $K_x(t) = \sum_{j=1}^J c_j \exp(-\lambda_j t)$. Introduce auxiliary states

$$d \mu_{ij}^x = [-\lambda_j \mu_{ij}^x + c_j x_i(t)] dt \quad (11)$$

$$M_i^x(t) \approx \sum_{j=1}^J \mu_{ij}^x(t) \quad (12)$$

The augmented process is Markov even though the projected dynamics of x_i are path dependent. Delayed extraction-displacement cycles can additionally use explicit lag τ_{ij} .

5.3 Hysteresis

Entry into and exit from vulnerability need not use the same threshold. Define $\psi_{\text{in}} > \psi_{\text{out}}$. This produces hysteresis and prevents instantaneous regime chattering:

$$Q \rightarrow V \text{ may occur when } \Psi_i \geq \psi_{\text{in}}; \quad V \rightarrow Q \text{ may occur only when } \Psi_i \leq \psi_{\text{out}} \quad (13)$$

6. Adaptive Topology and Higher-Order Interactions

6.1 Pairwise edge dynamics

At level k , $W^{(k)}(t) = [w_{ij}^{(k)}(t)]$ is a directed weighted network. Edge weights evolve in latent logistic coordinates y_{ij}^w , with $w_{ij} = \sigma(y_{ij}^w)$, so weights remain in $(0,1)$:

$$d y_{ij}^w = [\eta_U U_{ij} - \eta_K K_{ij} - \eta_X X_{ij} - \eta_H H_{ji} + \eta_S \text{Sim}_{ij} + \eta_M \text{Mem}_{ij}] dt + \sigma_w dB_{ij} \quad (14)$$

The terms represent reinforcing exchange, boundary enforcement, harmful extraction, harmful displacement, informational similarity, and relationship memory.

6.2 Edge birth and death

Potential edges absent from the active graph are created by a counting process with intensity

$$\lambda_{ij}^{\text{birth}} = \bar{\lambda}_b \sigma(k_b[\text{Sim}_{ij} + \text{Need}_{ij} + \text{CoLoc}_{ij} - \theta_b]) \quad (15)$$

Active edges disappear with intensity

$$\lambda_{ij}^{\text{death}} = \bar{\lambda}_d \sigma(k_d[\text{K}_{ij} + \text{Harm}_{ij} - \text{Benefit}_{ij} - \theta_d]) \quad (16)$$

6.3 Hyperedges and coalitions

Pairwise networks cannot represent coalition-level effects. Let $h_S(t)$ be the weight of a hyperedge connecting subset $S \subset V$. Its dynamics are

$$d y_S^h = [\eta_C \text{Cohesion}_S - \eta_F \text{Friction}_S - \eta_R \text{ResourceCost}_S] dt + \sigma_h dB_S, \quad h_S = \sigma(y_S^h) \quad (17)$$

Higher-order support or cascade pressure may then depend on $\sum_{S \ni i} h_S$ rather than only $\sum_j w_{ij}$.

6.4 Node split and merge

A system may split when internal modularity becomes persistent and cross-module coupling collapses. Systems i and j may merge when mutual information, alignment, and reciprocal support remain high over a dwell time. These events are represented by state-dependent hazards:

$$\lambda_i^{\text{split}} = \bar{\lambda}_s \sigma(k_s[\text{Mod}_i + s_i + \ell_i - \theta_s]) \quad (18)$$

$$\lambda_{ij}^{\text{merge}} = \bar{\lambda}_m \sigma(k_m[\text{MI}_{ij} + \text{Recip}_{ij} + \text{Align}_{ij} - \theta_m]) \quad (19)$$

7. Recursive Coarse-Graining and Downward Constraint

7.1 Membership and aggregation

Let $C^{(k)}$ be a column-stochastic membership matrix mapping level k systems into level $k+1$ systems. $C_{\{Ii\}}^{(k)}$ is the membership weight of lower-level system i in higher-level system I . The higher-level operational distribution is the information barycenter

$$P_I^{(k+1)} = \operatorname{argmin}_{\{P \in \Delta\}} \sum_i C_{Ii}^{(k)} D_{\text{geo}}(P_i^{(k)}, P) \quad (20)$$

where D_{geo} may be Jensen-Shannon, Fisher-Rao squared distance, or a Wasserstein cost. Official and meta-distributions are aggregated analogously, possibly with distinct weights.

7.2 Aggregated scalar states

For scalar states, define weighted aggregates plus interaction corrections:

$$e_I^{(k+1)} = \sum_i C_{Ii} e_i - \text{Cost}_I + \text{Synergy}_I \quad (21)$$

$$s_I^{(k+1)} = \sum_i C_{Ii} s_i + \text{Var}_C(P_i) + \text{CrossConflict}_I \quad (22)$$

$$f_I^{(k+1)} = \sigma(\beta_0 + \sum_i C_{Ii} \text{logit}(f_i) - \beta_s s_I) \quad (23)$$

7.3 Downward constraints

Higher-level systems feed back through normative, resource, and topological constraints. Let $\Lambda_i^{(k \leftarrow k+1)}$ denote the downward field:

$$\Lambda_i = C_{Ii} [B_O(O_{I-P_i}) + B_R \text{Resource}_I + B_K \text{Constraint}_I] \quad (24)$$

This term enters lower-level signal, energy, and edge equations. The hierarchy is therefore recursive rather than merely aggregated: lower levels construct higher levels, while higher levels alter lower-level trajectories.

7.4 Recursive closure

The same state definitions and evolution laws apply at every level k , with level-specific parameters. This is the formal recursive principle:

$$X^{(k+1)} = A_k(X^{(k)}, W^{(k)}, H^{(k)}, C^{(k)}); \quad dX^{(k)} \text{ depends on } X^{(k+1)} \quad (25)$$

8. Continuous Stochastic Dynamics

8.1 Probability-valued signals

Signals evolve on the simplex using a stochastic replicator/log-ratio formulation. Let $z_{Oa} = \log(O_a/O_m)$ for $a=1, \dots, m-1$, and similarly for P and M . The logistic-normal map returns valid distributions. In latent coordinates:

$$dz_{Oi} = [\kappa_O \text{Logit}(\Theta_i) - \kappa_O z_{Oi} + B_O \zeta_i + \Lambda_{Oi}]dt + \Sigma_O dW_{Oi} \quad (26)$$

$$dz_{Pi} = [\kappa_P (\text{Logit}(C_i) - z_{Pi}) - B_{\ell\ell_i} - B_{rr_i} + \text{Net}_i^P + \Lambda_{Pi} + \text{Mem}_i^P]dt + \Sigma_P dW_{Pi} \quad (27)$$

$$dz_{Mi} = [\kappa_M f_i (\text{Logit}(P_i) - \text{Logit}(O_i) - z_{Mi}) - \kappa_C (1 - f_i) z_{Mi} + \text{Mem}_i^M]dt + \Sigma_M dW_{Mi} \quad (28)$$

Θ_i is a retained normative target, C_i an incentive-compatible operational target, and Net_i^P a network transport term. Working in log-ratio coordinates guarantees that O_i , P_i , and M_i remain valid probability distributions.

8.2 Feedback permeability

$$dy_i^f = [\beta_{f0} + \beta_{fG} G_i^{\text{info}} + \beta_{fU} U_i - \beta_{f\ell\ell_i} - \beta_{fr} r_i - \beta_{fH} H_i - \beta_{fd} d_i + \beta_{f\mu} M_i^f]dt + \sigma_f dW_i^f \quad (29)$$

8.3 Fragmentation and stress

$$ds_i = [\alpha_{sd} d_i + \alpha_{sr} r_i + \alpha_{se}(e^* - e_i) + \alpha_{sH} H_i + \alpha_{sV} \text{Var}_i - \beta_{sf} f_i s_i - \beta_{sU} U_i s_i + \alpha_{s\mu} M_i^s]dt + \sigma_s \sqrt{s_i} dW_i^s \quad (30)$$

$$d\ell_i = [\alpha_{\ell d} d_i + \alpha_{\ell s} s_i + \alpha_{\ell H} H_i - \beta_{\ell f} f_i \ell_i - \beta_{\ell a} a_i \ell_i + \alpha_{\ell \mu} M_i^{\ell}]dt + \sigma_{\ell} \sqrt{\ell_i} dW_i^{\ell} \quad (31)$$

8.4 Usable reserve

$$de_i = [\pi_i A_i a_i + U_i^e - \mu_e e_i - c_{ss_i} - c_{\ell\ell_i} - X_i + \beta_{eI} I_i^{\text{pred}} + \Lambda_{ei}]dt + \sigma_e \sqrt{e_i} dW_i^e \quad (32)$$

$$dW_i^e$$

8.5 Rigidity and adaptive capacity

$$d y_i^r = [\beta_{r0} + \beta_{r\ell_i} + \beta_{rd} d_i + \beta_{rH} H_i - \beta_{rf} f_i - \beta_{rU} U_i + \beta_{r\mu} M_i^r]dt + \sigma_r dW_i^r \quad (33)$$

$$d y_i^a = [\beta_{a0} + \beta_{af} f_i + \beta_{ae} \log(1+e_i) + \beta_{aU} U_i + \beta_{aI} G_i^{\text{info}} - \beta_{as} s_i - \beta_{ar} r_i - \beta_{a\ell_i} + \beta_{a\mu} M_i^a]dt + \sigma_a dW_i^a \quad (34)$$

Bounded coordinates are recovered by $f_i = \sigma(y_i^f)$, $r_i = \sigma(y_i^r)$, and $a_i = \sigma(y_i^a)$. For exact analytical nonnegativity of s_i , ℓ_i , and e_i , use unregularized CIR-type diffusions satisfying appropriate boundary drift conditions; for numerical simulation, full truncation is acceptable and must be reported.

9. Extraction-Displacement-Support Coupling

For an edge $j \rightarrow i$, separate resource extraction, destabilizing displacement, and nonextractive support:

$$x_{ij} = w_{ji} \chi_x g_x(r_j, 1-f_j, \ell_j) g_v(e_i) \quad (35)$$

$$h_{ji}(t) = \rho_h x_{ij}(t-\tau_{ij}) + \chi_h w_{ij} g_h(\ell_j, d_j, r_j)[1-b_i] + \xi_{ji}(t) \quad (36)$$

$$u_{ji} = w_{ij} \chi_u a_j f_j b_i \cdot \exp[-d_{FR}(P_i, P_j)^2/\tau_u] \quad (37)$$

Total extraction, displacement, and support are $X_i = \sum_j x_{ij}$, $H_i = \sum_j h_{ji}$, and $U_i = \sum_j u_{ji}$ plus hyperedge contributions. The delayed return term in h_{ji} formalizes the path-dependent extraction-displacement cycle. A temporary zero in $d\ell_i/dt$ is not treated as equilibrium; local stability is determined by the memory-augmented Jacobian or characteristic equation.

10. State Regions and Hybrid Events

Define a composite risk functional that includes informational, structural, energetic, and historical terms:

$$\Psi_i = \omega_d d_i + \omega_s s_i + \omega_{\ell_i} \ell_i + \omega_r r_i - \omega_f f_i - \omega_a a_i - \omega_e \log(1+e_i) + \omega_{\mu} R_i^{\text{mem}} \quad (38)$$

where R_i^{mem} is a weighted memory burden. Quiescent and vulnerable regions use hysteretic boundaries $\psi_{\text{out}} < \psi_{\text{in}} < \psi_C$:

$$Q = \{X: \Psi_i \leq \psi_{\text{out}}\}; \quad V = \{X: \psi_{\text{in}} < \Psi_i < \psi_C\}; \quad B = \{X: \psi_{\text{out}} < \Psi_i \leq \psi_{\text{in}}\} \quad (39)$$

B is a buffer region that preserves the previous regime. Discrete regimes are $R_i \in \{Q, V, C, D, R\}$, representing quiescent operation, vulnerability, functional collapse, disintegration, and restoration trajectory.

10.1 Jump intensities

$$\lambda_i^{\wedge}(Q \rightarrow V) = \bar{\lambda}_{QV} \sigma(k_{QV}[\Psi_i - \psi_{\text{in}}]) \quad (40)$$

$$\lambda_i^{\wedge}(V \rightarrow C) = \bar{\lambda}_{VC} \sigma(k_{VC}[\Psi_i - \psi_C + \gamma_C C_i^{\text{net}} - \gamma_Y Y_i]) \quad (41)$$

$$\lambda_i^{\wedge}(V \rightarrow D) = \bar{\lambda}_{VD} \sigma(k_{VD}[s_i - s_D + \gamma_D \ell_i]) \quad (42)$$

$$\lambda_i^{\wedge}(V \rightarrow R) = \bar{\lambda}_{VR} \sigma(k_{VR}[Y_i - Y_C + e_i - e_C + f_i - f_C]) \quad (43)$$

$$\lambda_i^{\wedge}(V \rightarrow Q) = \bar{\lambda}_{VQ} \sigma(k_{VQ}[\psi_{out} - \Psi_i + \gamma_U U_i]) \quad (44)$$

$$\lambda_i^{\wedge}(C \rightarrow R) = \bar{\lambda}_{CR} \sigma(k_{CR}[U_i - u_C + f_i - f_C]) \quad (45)$$

$$\lambda_i^{\wedge}(R \rightarrow Q) = \bar{\lambda}_{RQ} \sigma(k_{RQ}[A_i - A_R + a_i - a_R - \Psi_i + \psi_{out}]) \quad (46)$$

The yielding index now depends on adaptive capacity, permeability, reserve, and state-space speed:

$$Y_i = a_i f_i [e_i / (e_i + e_C)] \cdot \sigma(k_v[v_i - v_C]) \quad (47)$$

Network cascade pressure includes failed neighbors, hyperedges, and higher-level failure:

$$C_i^{\wedge net} = \sum_j w_{ij} 1_{\{R_j \in \{C, D\}\}} + \kappa_H \sum_j h_{ji} + \sum_{\{S \ni i\}} h_S \text{Fail}_S + \kappa_{\uparrow} \text{Fail}_{parent(i)} \quad (48)$$

11. Reset, Split, Merge, and Hierarchy Maps

Each jump applies a measurable reset map J . Illustrative maps are:

Event	Reset / structural map
$V \rightarrow C$	$e \leftarrow \rho_C e$; $f \leftarrow \rho_f f$; $a \leftarrow \rho_a a$; $r \leftarrow \min(1, r + \Delta r_C)$
$V \rightarrow D$	$s \leftarrow s_D$; disable or reassign incident edges; invalidate local aggregation map
$C \rightarrow R$	raise minimum permeability; activate external support; attenuate harmful edges
$R \rightarrow Q$	no discontinuous reset; require alignment and performance dwell condition
edge death	remove edge and associated delayed-memory state
node split	partition state by module weights; divide reserve; reconstruct child distributions and edges
node merge	form information barycenter; pool reserve minus integration cost; combine memories with weights

hierarchy birth	create parent node from persistent coalition/hyperedge and install membership map C
hierarchy dissolution	remove parent node and release or remap downward constraints

11.1 Restoration criterion

Restoration is not achieved by redefining the official signal to match dysfunction. Let Θ_i be fixed or externally constrained over the restoration horizon. Restoration succeeds when, for dwell time T_R ,

$$d_i(t) \leq d_R, f_i(t) \geq f_R, a_i(t) \geq a_R, \Psi_i(t) \leq \psi_{out}, \text{ and } J_i(P_i(t)) \geq J_R \quad (49)$$

for all t in $[\tau, \tau + T_R]$. J_i is an external performance functional, preventing trivial relabeling.

12. Generator and Well-Posedness

After adding the finite memory coordinates, the full state is a piecewise-diffusion Markov process with variable dimension. Between structural jumps, for smooth test function φ , the generator is

$$L\varphi = \nabla\varphi \cdot b(z) + 1/2 \text{Tr}[\Sigma\Sigma^T \nabla^2\varphi] + \sum_e \lambda_e(z)[\varphi(J_e z) - \varphi(z)] \quad (50)$$

The event index e ranges over regime transitions, edge birth/death, hyperedge birth/death, node split/merge, and hierarchy formation/dissolution. Variable-dimension jumps are defined on the disjoint union of state spaces over all admissible graph sizes.

Sufficient conditions for local existence and nonexplosion include locally Lipschitz latent-coordinate drifts, linear growth bounds, finite total hazard on compact sets, non-Zeno dwell or hazard conditions for structural events, and measurable reset maps that preserve admissible distributions and bounded coordinates.

13. Stability, Criticality, and Cascade Criteria

13.1 Lyapunov functional with memory

A candidate multiscale Lyapunov functional is

$$V(Z) = \sum_k \rho_k \sum_i [c_d d_i + c_s s_i + c_{\ell} \ell_i + c_r r_i^2 + c_e (e_i - e^*)^2 - c_f \log f_i - c_a \log a_i + c_\mu \|\mu_i\|^2] + \frac{c_W}{\|W - W^*\|^2} \quad (51)$$

If $LV \leq -cV + K$ outside a compact aligned neighborhood and jump resets satisfy $E[V(J_e Z) - V(Z) | Z] \leq -\delta_e V + K_e$ for restorative events, the process is stochastically bounded and recurrent toward the aligned set under standard irreducibility assumptions.

13.2 Memory-induced instability

Linearization of the Volterra system around equilibrium yields a characteristic equation

$$\det[\lambda I - A_0 - \hat{K}(\lambda)A_1] = 0 \quad (52)$$

where $\hat{K}$ is the Laplace transform of the memory kernel. Instability occurs when a root crosses into the right half-plane. Thus long memory can destabilize an equilibrium even when the memoryless Jacobian A_0 is stable.

13.3 Network and hierarchy criticality

Let G_{eff} be the linearized gain matrix across pairwise, hyperedge, and downward channels. A sufficient local cascade condition is

$$\rho(G_{\text{eff}}) > 1 \quad (53)$$

where ρ denotes spectral radius. For time-varying networks, use the top Lyapunov exponent of the random matrix cocycle. Hierarchical failure can amplify or damp lower-level cascades depending on the sign and gain of downward constraints.

14. Identifiability and Empirical Mapping

The model is modular. Applications need not observe every state directly. Probability-valued signals can be estimated from empirical frequency distributions, classifier outputs, language-model embeddings converted to categorical distributions, policy coding, event logs, or behavioral sequences. Latent states can be inferred using particle MCMC, sequential Monte Carlo, variational filtering, or simulation-based inference.

Latent object	Possible proxies
O,P,M distributions	declared policies, observed action frequencies, audit/self-model outputs
JS / Fisher-Rao divergence	distributional discrepancy across declared, realized, and monitored behavior
feedback permeability	correction latency, anomaly acknowledgment, audit access, learning rate
memory burden	incident backlog age distribution, unresolved exceptions, physiological or technical hysteresis
adaptive topology	communication links, transaction networks, module dependencies, coalition membership
hierarchy map C	organizational membership, subsystem composition, multiscale clustering
predictive information	forecast accuracy, mutual information between present and future outputs
fragmentation	cross-module inconsistency, modular disagreement, coordination failures
usable reserve	compute, liquidity, staffing, metabolic reserve,

	slack, attention
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Structural identifiability must be checked before empirical estimation. In particular, divergence, permeability, and latent stress may be confounded unless the observation model includes independent error-response and performance measurements. Hierarchical aggregation parameters are identifiable only when lower-level and higher-level observations overlap in time.

15. Simulation-Ready Specification

- Choose K levels, initial node sets, membership matrices, pairwise networks, and hyperedges.
- Represent O , P , and M in additive-log-ratio coordinates; represent f , r , a , w , and h in logistic coordinates.
- Approximate memory kernels with J exponential modes and initialize auxiliary memory states.
- At each time step, compute information divergences, surprise, predictive information, network flows, hyperedge flows, and interlevel constraints.
- Advance continuous latent coordinates with Euler-Maruyama, Milstein, or a positivity-preserving scheme.
- Sample competing regime, topology, split/merge, and hierarchy events from their integrated hazards.
- Apply reset maps and rebuild graph, memory, and hierarchy objects after variable-dimension events.
- Recompute higher-level barycenters and downward fields.
- Record stopping times, cascade sizes, topology changes, hierarchy changes, and information-flow statistics.
- Perform sensitivity analysis over time step, kernel approximation, network density, hierarchy depth, thresholds, and random seed.

15.1 Compact pseudocode

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initialize multiscale state  $Z$ , graph  $W$ , hypergraph  $H$ , hierarchy  $C$ , memory  $\mu$ 
for  $t = 0, \dots, T-\Delta t$ :
  map latent signal coordinates to  $O, P, M$  on simplex
  compute JS, Fisher-Rao distance, surprise, predictive information
  update memory modes  $\mu$  and delayed channel buffers
  compute extraction, displacement, support, pairwise and hyperedge influence
  aggregate upward states and compute downward constraints
  advance continuous SDEs
  compute all active event hazards
  sample competing regime/topology/split/merge/hierarchy events
  apply reset and variable-dimension maps
  rebuild affected networks, memories, and membership matrices
  record trajectories and stopping times

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16. Core Testable Predictions

- After controlling for current output, collapse hazard increases with information divergence, fragmentation, latent stress, rigidity, harmful memory burden, and cascade pressure, and decreases with permeability, adaptive capacity, reserve, and predictive information.
- Externally stable official distributions do not imply low risk when operational and meta-distributions diverge and feedback permeability is low.
- Long-memory kernels produce hysteresis, delayed collapse, and recovery asymmetry that a memoryless model cannot reproduce.

- Adaptive rewiring can reduce local risk while increasing system-wide modular fragmentation or transferring risk to a higher level.
- Coalition or hyperedge effects can trigger cascades even when every pairwise coupling is individually subcritical.
- Recursive aggregation can hide lower-level instability when barycentric averaging cancels opposing deviations; variance and cross-conflict terms recover this hidden risk.
- Downward constraints can stabilize constituents when aligned and nonextractive, but amplify collapse when higher-level divergence and rigidity are high.
- Large perturbations produce restoration rather than collapse when yielding, reserve, permeability, and information assimilation remain above threshold.
- Split events are favored by persistent internal modularity and fragmentation; merge events are favored by high mutual information, reciprocal support, and alignment.
- Relabeling declared objectives can reduce measured divergence but will not satisfy restoration when the retained normative target and external performance functional are fixed.

17. Compact Mathematical Formulation

$$dX = b(X, R, W, H, C, \mu, u, t)dt + \Sigma(X, R, W, H, C, t)dB_t \quad (54)$$

$$d\mu = A_\mu \mu dt + B_\mu X dt \quad (55)$$

$$P(dN_e=1 | F_t) = \lambda_e(Z_t)dt + o(dt) \quad (56)$$

$$Z \leftarrow J_e(Z) \text{ when } dN_e=1 \quad (57)$$

$$dY_W = g_W(Z)dt + \Sigma_W dB_W, \quad W = \sigma(Y_W) \quad (58)$$

$$dY_H = g_H(Z)dt + \Sigma_H dB_H, \quad H = \sigma(Y_H) \quad (59)$$

$$X^{(k+1)} = A_k(X^{(k)}, W^{(k)}, H^{(k)}, C^{(k)}); \quad dX^{(k)} \text{ includes } D_k(X^{(k+1)}) \quad (60)$$

Equations (54)-(60), together with information-geometric signal definitions, memory embedding, state-dependent event hazards, and variable-dimension reset maps, define the expanded RICS-FDFM model.

18. Conclusion

RICS-FDFM is properly represented as a recursive multiscale information-geometric stochastic hybrid system on an adaptive network and hypergraph. Its information variables are explicit probability distributions; its signal discrepancies are geometric and information-theoretic; its topology can reorganize; its hierarchy is generated recursively and feeds back downward; and its dynamics retain history through memory kernels. The model preserves the original conceptual architecture while replacing metaphorical primitives with measurable states, stochastic flows, adaptive graph processes, recursive aggregation, stopping times, jump hazards, and stability criteria. It is sufficiently explicit to simulate, estimate, compare against reduced models, and falsify.

Appendix A. Suggested Default Functions

$$\sigma(z)=1/(1+e^{-z}); \text{ softplus}(z)=\log(1+e^z); (z)_+=\max(z,0) \quad (\text{A1})$$

$$g_x(r,1-f,\ell)=\text{softplus}(\kappa_x0+\kappa_{xr}r+\kappa_{xf}(1-f)+\kappa_{x\ell}\ell) \quad (\text{A2})$$

$$g_h(\ell,d,r)=\text{softplus}(\kappa_h0+\kappa_{h\ell}\ell+\kappa_{hd}d+\kappa_{hr}r) \quad (\text{A3})$$

$$g_v(e)=e/(e+e_0) \quad (\text{A4})$$

$$K_x(t)=\sum_j c_j \exp(-\lambda_j t), \quad c_j \geq 0, \lambda_j > 0 \quad (\text{A5})$$

These functions are placeholders for application-specific response laws, not universal causal laws.

Appendix B. Numerical Safeguards

- Use additive-log-ratio, centered-log-ratio, or softmax coordinates for probability-valued signals.
- Use latent logistic coordinates for all variables constrained to (0,1).
- For square-root diffusions, either preserve the exact boundary structure and verify Feller conditions or use full truncation and state clearly that positivity is numerical.
- Use adaptive time stepping near high hazards and after split/merge events.
- Approximate long-memory kernels with enough exponential modes to match the target kernel over the simulation horizon.
- Prevent Zeno behavior by imposing minimum dwell times or bounded integrated hazards for repeated structural events.
- After node split or merge, renormalize probability distributions and membership matrices and verify reserve conservation minus specified costs.
- Check sensitivity to hierarchy depth, hyperedge order, topology priors, thresholds, memory-kernel choice, time step, and random seed.
- Distinguish structural identifiability from empirical estimability before interpreting fitted parameters.

Appendix C. Core Symbol Table

Symbol	Definition
O,P,M	official, operational, and meta-signal probability distributions
d, A	information divergence and normalized alignment
f,s,ℓ,e,r,a	permeability, fragmentation, stress, reserve, rigidity, adaptability
μ	finite-dimensional memory embedding

W, H	pairwise network and hypergraph weights
C	interlevel membership/coarse-graining map
X, H, U	total extraction, displacement, and support flows
Ψ	composite risk functional
R	discrete regime variable
J_e	reset or structural map for event e
λ_e	state-dependent event intensity
A_k, D_k	upward aggregation and downward-constraint operators
I^{pred}	predictive mutual information
S_{rp}	Bayesian surprise
Y	yielding/survival index